Two-Dimensional Spectral Analysis Filter for Removal of LFM Radar Interference in Spaceborne SAR Imagery

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***Abstract*— Radio spectrum bands allocated to spaceborne synthetic aperture radar (SAR) imagery are shared by multiple missions. In practical radio spectrum environments, these bands are also used by some ground radars, e.g., C-band weather radar. Due to this fact, radio frequency interference (RFI) may occur for a spaceborne SAR when its received signals contain the transmitted waveforms from another SAR or radar operating at the same frequency band. This particular class of RFI is usually linear-frequency-modulation (LFM) signals, which can cause bright radiometric artifacts in focused SAR images. Most existing signal processing approaches designed for addressing this problem belong to the class of preprocessing methods, which removes RFI in level-0 raw radar data before SAR focusing. In this article, we propose a postprocessing kernel—2-D SPECtral ANalysis (2-D SPECAN) filter, for removing the class of LFM RFI in level-1 SLC images. The filtering consists of three main steps: Step 1: focus LFM RFI artifacts in SLC images as point-like responses in the spectral domain via 2-D SPECAN; Step 2: perform 2-D notch filtering in the spectral domain to remove the most contribution of the RFI responses; and Step 3: transform the filtered spectrum back into the SLC image domain using the inverse operation of the 2-D SPECAN. For computation efficiency, we design a simplified processing flow and adopt a blockwise processing strategy. Experiments with several Sentinel-1 SLC images demonstrate that severe RFI artifacts in SLC images can be removed significantly by the proposed method.**

***Index Terms*— Radio frequency interference (RFI), Sentinel-1, SPECtral ANalysis (SPECAN), synthetic aperture radar (SAR).**

I. Introduction

A. Background

As WE all know, synthetic aperture radar (SAR) works by transmitting radio signals and receiving target echoes in a certain radio spectrum band. Since the spectrum is shared by many different radio systems, SAR may also receive the signals from other radio devices while receiving the target echoes, causing the potential occurrence of radio frequency interference (RFI). In order to avoid RFI as much as possible, the International Telecommunication Union (ITU) has proposed some regulations and recommendations on the management and allocation [1] of radio spectrum resources. However, RFI is still a common problem in spaceborne SAR Earth observation due to limited spectrum resources, especially in low-frequency bands.

Low-frequency RFI in spaceborne SAR usually comes from AM/FM broadcasting, television, and communication networks, which usually do not exist in the C-band. In this band, RFI due to transmitted signals from ground-based radars is common. In addition, two SAR satellites can have mutual interference to each other in some occasional cases [2]–[5], e.g., when one SAR satellite falls into the main beam lobe of another SAR satellite. This class of RFI signals is linear-frequency-modulation (LFM) pulses, which are widely used in radar to obtain resolving ability in the range direction. The consequence of the pulsed LFM RFI is that they can cause radiometric artifacts (which look like bright strips) in SAR images, contaminating or even submerging informative image data, as shown by the Sentinel-1 quick-look image in Fig. 1. Consequently, the presence of RFI will bring difficulties to subsequent signal processing and applications, such as SAR interferometry and SAR polarimetry [6], [7].

B. Related Works

Over the past decades, many signal processing methods have been proposed for suppressing RFI in SAR raw data, e.g., notch filtering [8], eigen subspace projection (ESP) [9], [10], time–frequency filtering [11], spectrum estimation [12], [13], deep leaning [14], and sparse optimization and low-rank matrix restoration [15]–[17], [17]–[25]. Notch filtering discards the spectrum polluted by narrowband RFI in the range spectrum domain. ESP constructs a subspace filter for each pulse data in the range time domain to remove narrowband RFI. Time–frequency filtering extends the conventional notch filtering into range time–frequency domain to tackle nonstationary wideband RFI. Sparse spectrum estimation methods use the sparse property of narrowband RFI to estimate and subtract RFI from SAR data. Low-rank matrix restoration methods utilize the low-rank property of narrowband RFI to obtain performance improvements over

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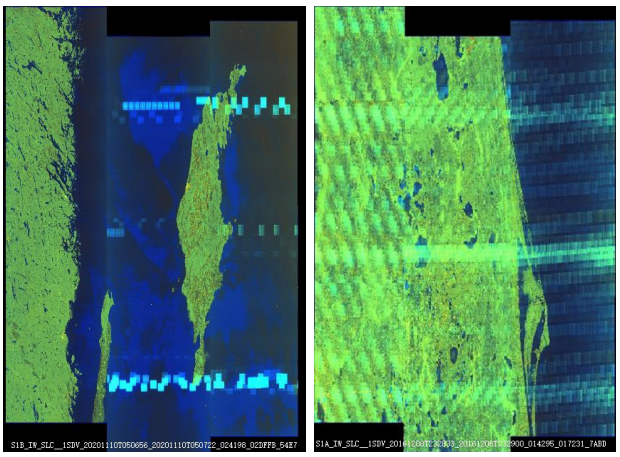

Fig. 1. Sentinel-1 quick look images with radiometric artifacts caused by pulsed LFM RFI.

conventional methods, at the cost of significantly increased computation burden due to involving iterative optimization and singular value decomposition (SVD).

A common point between these methods is that they act as preprocessing methods applied to SAR level-0 data. However, due to the fact that RFI is overall not a global problem in spaceborne SAR Earth observation, e.g., in the C-band, RFI filtering on SAR raw data is not widely used. As a consequence, some level-1 single-look-complex (SLC) image products of SAR satellites, e.g., Sentinel-1, Radarsat-2, and Gaofen-3, contain noticeable RFI artifacts. Although the RFI problem is, in general, not globally frequent for these sensors, it, indeed, frequently happens over some particular observation areas, such as the Baltic Sea in North Europe, as two examples shown in Fig. 1. Since SAR data accessible to users are usually level-1 or higher, it is important to design technology for removing radar interference signatures in the image domain, which is applied merely on SLC images and does not require raw data processing.

Another class of RFI removal techniques for SAR works as postprocessing methods, which suppresses RFI after SAR focusing. In [26], an inverse operation of SAR focusing is used to transform focused SLC images into the azimuth-time range-frequency domain, followed by the conventional notch filtering to remove L-band RFI. The drawback of this method is that the processing steps are complicated. Recently, a robust principal component analysis (PCA) is proposed in [5] to remove RFI artifacts due to LFM signals in SLC SAR images directly. This method, unlike most existing ones that belong to the class of radar signal processing approaches, is an image processing method. However, it has a limitation of relying on the assumption of Gaussian reflectivity and high interference-to-signal ratio to ensure the accurate extraction of RFI components via PCA. Robust PCA can partly reduce the statistical requirement on the reflectivity of the observed region, as reported in [5]. However, similar to low-rank matrix restoration methods, it has a large computational cost due to involving iterative optimization and SVD.

C. Research Scope and Contributions of This Article

In this article, we are concerned with a particular class of RFI in SAR imagery, i.e., pulsed LFM RFI usually emitted by ground radars or another SAR satellites. The occurrences of this class of RFI are not globally frequent in SAR Earth observation but are very common for some particular regions, i.e., North Europe. The goal of this article is to design an efficient postprocessing method for removing the radiometric artifacts caused by this class of RFI in SLC SAR images.

To this end, we exploit the signal property of the RFI artifacts to develop a three-step method. The first step is to focus the RFI artifacts, which have wide azimuth and range extents, into narrow sinc responses via a 2-D SPECtral ANalysis (SPECAN) method. SPECAN is an algorithm originally used for SAR azimuth focusing on exploiting the LFM property of SAR azimuth signals [27], [28]. Different from the original SPECAN, here 2-D SPECAN refers to a method using SPECAN operation in both azimuth and range to focus 2-D LFM signals. According to [5], the RFI artifact caused by a pulsed LFM waveform is approximately a 2-D LFM signal, and thus, it is feasible to focus the RFI artifact via the 2-D SPECAN method. Because a Fourier transform is involved in the RFI focusing, we call the RFI-focused data domain the 2-D spectral domain. The second step is to construct notches in the 2-D spectral domain to notch out the focused point-like responses of the RFI artifacts. This is realized using a thresholding operation and an RFI support augmentation operation, which generates notches covering the focused RFI responses. In the third step, we transform the processed data back into the original image domain using steps inverse to the 2-D SPECAN. Since SLC SAR images typically have very large sizes and are difficult to apply the proposed method directly, we use a blockwise processing strategy, which takes the proposed method as its processing kernel.

The remainder of this article is organized as follows. Section II introduces the signal model of pulsed LFM RFI. Section III presents theoretic basics for the proposed method, i.e., RFI focusing via 2-D SPECAN processing. Section IV designs a fast 2-D notch filtering method for the removal of focused RFI responses in the spectral domain. Section V presents a simplified processing flow for simplifying discrete signal processing. Section VI provides several experiments using Sentinel-1 SLC images polluted by real pulsed LFM RFI caused by radar. Section VII concludes this article.

II. Signal Model

This section introduces the signal model of RFI artifacts caused by radar LFM pulses. Following conventions in the literature of SAR signal processing, we will describe SAR images with continuous signal forms. The actual SAR images are the discrete samples of their associated continuous signals.

A. Signal Model of Radar Interference in SAR Echoes

In SAR imagery, RFI happens if there exist other radio signals received by an SAR along with the echo backscattered from the SAR illuminated region. Considering the presence of RFI and observation noise, the received SAR signal is the superposition of three terms as follows:

$$s(\eta, t) = x(\eta, t) + r(\eta, t) + z(\eta, t) \tag{1}$$

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where η and t are the slow and fast time, respectively, and $x(\eta, t)$, $r(\eta, t)$, and $z(\eta, t)$ are the signal terms of backscattered echo, RFI, and observation noise, respectively.

There are many potential radio applications that generate RFI signals for SAR, such as radio broadcasting, communications, navigation, radar, space exploration, and amateur radio. In this article, we study the case of radar interference. In this case, the signal form of RFI depends on the radar waveform type. In SAR and radar, LFM is the mostly used waveform, which is widely used in various ranging and imaging radars. As observed in Sentinel-1 imagery, LFM RFI is very common, especially for certain regions.

Based on the above considerations, we consider the LFM signal model for radar interference

$$p(t) = \text{rect}\left(\frac{t}{T_i}\right)e^{j2\pi f_c t + j\pi K_i t^2} \tag{2}$$

where rect$(\cdot)$ is the rectangular function, f_c is the center frequency of the RFI subtracting the SAR center frequency, and T_i and K_i are the pulse duration and FM rate of the RFI, respectively. The subscript "i" means "interference." Using the above 1-D LFM signal model, we can express the RFI signal consisting of L LFM pulses in the 2-D echo domain as

$$r(\eta, t) = \sum_{l=1}^{L} a_l \delta(\eta - \eta_l) p(t - t_l) \tag{3}$$

where a_l is a complex number whose amplitude is relating to the RFI power level, $\delta(\eta)$ is the Dirac function, η_l is the slow time that embeds the lth RFI pulse, and t_l is the propagation delay of the lth RFI pulse, respectively. The above signal model can describe most occurrences of RFI caused by ground radar and another SAR satellite. In fact, there exist other signal forms for SAR and radar, for example, phase-coded signals, nonlinear frequency-modulated chirps, and waveforms designed for MIMO SAR and radar [29]–[33]. However, we have not find occurrences of RFI due to these signals, so we do not consider these signal forms in this article.

B. Signal Model of Radar Interference in SLC SAR Images

Since, in essence, SAR focusing is a 2-D matched filtering process, we can express the focused image using the continuous signal

$$\begin{aligned}\tilde{s}(\eta', t') &= s(\eta, t) * h(\eta, t; \eta', t') \\ &= \tilde{x}(\eta', t') + \tilde{r}(\eta', t') + \tilde{z}(\eta', t')\end{aligned} \tag{4}$$

where η' and t' are the azimuth time and range time in the focused image domain, $h(\eta, t; \eta', t')$ is a time-varying convolution kernel, and the last three terms represent the focused image signal, the image-domain response of the LFM RFI, and the noise contribution, respectively. The response of LFM RFI can be written as $\tilde{r}(\eta', t') = \sum_{l=1}^{L} \tilde{r}_l(\eta', t')$ where

$$\tilde{r}_l(\eta', t') = a_l \delta(\eta - \eta_l) p(t - t_l) * h(\eta, t; \eta', t'). \tag{5}$$

In this article, we assume that the SAR system uses LFM pulses as its transmitting signals, based on which we can approximate the focusing kernel using a 2-D LFM signal

$$h(\eta, t) = \text{rect}\left(\frac{\eta}{T_a}\right)e^{-j\pi K_a \eta^2}\text{rect}\left(\frac{t}{T_r}\right)e^{-j\pi K_r t^2} \tag{6}$$

Based on this approximation and using the method of stationary phase, we can approximately express the image-domain response of the lth LFM RFI pulse as

$$\begin{aligned}\tilde{r}_l(\eta', t') &= a_l \delta(\eta - \eta_l) p(t - t_l) * h(\eta, t) \\ &= a_l \text{rect}\left(\frac{\eta' - \eta_l}{T_a}\right)e^{-j\pi K_a(\eta' - \eta_l)^2} \\ &\quad \times \text{rect}\left(\frac{t' - t_l}{T_r'}\right)e^{j2\pi f_c(t' - t_l) + j\pi K_r'(t' - t_l)^2}\end{aligned} \tag{7}$$

where $K_r' = (K_r K_i/(K_i - K_r))$ and $T_r' = T_i|K_i - K_r|/K_r$. Here, we assume that $K_i \neq K_r$, which is the common case in spaceborne SAR imagery. In the following, we will use the above parametric signal model of LFM interference artifact to develop an efficient method for RFI removal in SLC images.

III. 2-D SPECAN for Focusing RFI Artifacts

According to the signal model in (7), PCA and RPCA are introduced for RFI removal exploiting the low-rank property of RFI artifacts [5]. As popular techniques in image processing, the two methods are easy to implement and do not require specialized knowledge in SAR signal processing. However, they have some drawbacks. First, PCA does not preserve strong point scatterers. Second, both the two methods have a noticeable computational burden due to involving SVD. In the following, we will propose a new algorithm to avoid these drawbacks.

A. RFI Focusing via Matched Filtering

Before processing, we limit our problem to the case of single RFI source emitting pulsed LFM RFI with the same FM rate. From (7), we can see that the response of each LFM RFI pulse is the shifted and scaled version of

$$\tilde{r}_0(\eta', t') = \text{rect}\left(\frac{\eta'}{T_a}\right)e^{-j\pi K_a \eta'^2}\text{rect}\left(\frac{t'}{T_r'}\right)e^{j2\pi f_c t' + j\pi K_r' t'^2}. \tag{8}$$

Using this notation, we can express $\tilde{r}_l(\eta', t') = a_l \tilde{r}_0(\eta' - \eta_l, t' - t_l)$. This expression is similar to the basic delay and scaled echo model in radar. As we know in radar signal processing, matched filtering is used to compress the radar echo of a point target as a narrow sinc response, which also obtains a processing gain in signal-to-noise ratio. Similarly, we can use matched filtering to compress the RFI artifact as a narrow sinc response with an increased interference-to-signal ratio such that the RFI is less coupled with clean SAR data and, thus, can be easier to filter out in the match filtering output. Thus, we apply matched filtering to the focused image signal $\tilde{s}(\eta', t')$

$$\tilde{s}(\eta, t)_{\text{mf}} = \iint \tilde{s}(\eta', t')\tilde{r}_0^*(\eta' - \eta, t' - t)d\eta' dt' \tag{9}$$

where $\tilde{r}_0^*(-\eta', -t')$ is taken as the convolution kernel and $(\cdot)^*$ denotes the conjugate operation. Due to the linearity of matched filtering, the output signal $\tilde{s}(\eta, t)_{\text{mf}}$ can be expressed

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as the sum of the matched filtering output of the clean signal, the RFI, and the noise, respectively,

$$\tilde{s}(\eta,t)_{\mathrm{mf}}=\tilde{x}_{\mathrm{mf}}(\eta,t)+\tilde{r}_{\mathrm{mf}}(\eta,t)+\tilde{r}_{\mathrm{mf}}(\eta,t). \tag{10}$$

In this equation, $\tilde{r}_{\mathrm{mf}}(\eta,t)$ can be further written as the sum of L RFI terms as $\tilde{r}_{\mathrm{mf}}(\eta,t)=\sum_{l=1}^{L}\tilde{r}_{l,\mathrm{mf}}(\eta,t)$. The clean signal and noise signal terms, $\tilde{x}_{\mathrm{mf}}(\eta,t)$ and $\tilde{r}_{\mathrm{mf}}(\eta,t)$, are flat signals, as they are uncorrelated with the 2-D chirp signal $\tilde{r}_l(\eta',t')$. In other words, $\tilde{x}(\eta',t')$ and $\tilde{r}(\eta',t')$ are not accumulated in the matched filtering output, whereas $\tilde{r}_{\mathrm{mf}}(\eta,t)$ is accumulated as the sum of multiple sharp sinc functions centered at $(\eta,t)=(\eta_l,t_l)$, $l=1,\ldots,L$. The detailed derivation to explain this mechanism is given in the following equation:

$$\begin{aligned}
&\tilde{r}_{l,\mathrm{mf}}(\eta,t)\\
&=\iint\tilde{r}_l(\eta',t')\tilde{r}_0^*(\eta'-\eta,t'-t)d\eta'dt'\\
&=\int\tilde{R}(f_a,f_)\tilde{R}_0^*(f_a,f_r)e^{j2\pi f_a\eta+j2\pi f_rt}df_adf_r\\
&=a_lC\int\mathrm{rect}\left(\frac{f_a}{|K_a|T_a}\right)e^{j\pi\frac{f_a^2}{K_a}}e^{-j2\pi f_a\eta_l}\\
&\qquad\times\mathrm{rect}\left(\frac{f_a}{|K_a|T_a}\right)e^{-j\pi\frac{f_a^2}{K_a}}e^{2\pi f_a\eta}df_a\\
&\quad\int\mathrm{rect}\left(\frac{f_r-f}{|K_r'|T_r'}\right)e^{-j\pi\frac{(f_r-f)^2}{K_r'}}e^{-j2\pi f_rt_l}\\
&\qquad\times\mathrm{rect}\left(\frac{f_r-f}{|K_r'|T_r'}\right)e^{j\pi\frac{(f_r-f)^2}{K_r'}}df_re^{2\pi f_rt}\\
&=a_lC\int_{-\frac{|K_a|T_a}{2}}^{\frac{|K_a|T_a}{2}}e^{-j2\pi f_a\eta_l}df_a\int_{-\frac{|Kr'|T_r'}{2}}^{\frac{\left|K_r'\right|T_r'}{2}}e^{-j2\pi f_rt_l}df_r\\
&=a_lC|K_a|T_a\mathrm{sinc}(|K_a|T_a(\eta-\eta_l))\left|K_r'\right|T_r'\mathrm{sinc}\left(\left|K_r'\right|T_r'(t-t_l)\right).
\end{aligned} \tag{11}$$

Briefly speaking, this expression states that an image-domain LFM RFI artifact is focused as a point-like sinc response by the matched filtering operation. Let $B_a=|K_a|T_a$ and $B_r'=|K_r'|T_r'$; we can explicitly express the focused RFI response as

$$\tilde{r}_{\mathrm{mf}}(\eta,t)=\sum_{l=1}^{L}a_lB_a\mathrm{sinc}(B_a(\eta-\eta_l))B_r\mathrm{sinc}\left(B_r'(t-t_l)\right) \tag{12}$$

which is a combination of L narrow 2-D sinc responses. By suppressing these sinc responses in the matched filtering output, RFI can be effectively mitigated. Before introducing the suppressing method for the sinc responses of RFI, we introduce a computationally efficient algorithm for performing the matched filtering. Referring to (11), we can find that RFI can be focused in the matched filtering output via three steps.

1) Apply Fourier transform to $\tilde{s}(\eta,t)_{\mathrm{mf}}$ to obtain its spectrum $\tilde{S}(f_a,f_r)_{\mathrm{mf}}$.
2) Multiply the image spectrum $\tilde{S}(f_a,f_r)_{\mathrm{mf}}$ with the reference spectrum $\tilde{R}_0^*(f_a,f_r)$.
3) Apply inverse Fourier transform to obtained the matched filtering output.

In the discrete version of the above operations, it involves two 2-D FFT and one matrix Hadamard product (elementwise multiplication). Suppose that the image size is $N_a\times N_r$, the computational complexity of the 2-D FFT is $N_aN_r(\log N_r+\log N_a)$, and that of the matrix Hadamard product is N_aN_r. It is clear that the two 2-D FFT dominates the computational cost, and therefore, it is helpful to reduce the number of FFT operations.

B. Efficient RFI Focusing via 2-D SPECAN

In order to reduce the number of FFT operations, we adopt the SPECAN technique for computing the matched filtering. SPECAN is originally used in SAR as a 1-D focusing algorithm, which uses the quadratic property of SAR azimuth signals to perform azimuth compression by means of azimuth deramping and azimuth FFT. In this article, we extend the SPECAN as a 2-D algorithm for computing the matched filtering in (9). The mathematical development of the 2-D SPECAN is given by the following equation:

$$\begin{aligned}
\tilde{r}(\eta,t)_{l,\mathrm{mf}}&=\iint\tilde{r}_l(\eta',t')\mathrm{rect}\left(\frac{\eta'-\eta}{T_a}\right)e^{j\pi K_a(\eta'-\eta)^2}d\eta'\\
&\qquad\times\mathrm{rect}\left(\frac{t'-t}{T_r'}\right)e^{-j2\pi f_c(t'-t)-j\pi K_r'(t'-t)^2}dt'\\
&\approx\iint\tilde{r}_l(\eta',t')e^{j\pi K_a(\eta'-\eta)^2}d\eta'\\
&\qquad\times e^{-j2\pi f_c(t'-t)-j\pi K_r'(t'-t)^2}dt'\\
&=e^{-j\pi K_r't^2+j2\pi f_ct}\\
&\quad\times\int e^{j\pi K_a\eta^2}\int\tilde{r}_l(\eta',t')e^{j\pi K_a\eta'^2}e^{-j2\pi K_a\eta\eta'}d\eta'\\
&\qquad\times e^{-j\pi K_r't'^2}e^{j2\pi(K_r't-f)t'}dt'.
\end{aligned} \tag{13}$$

The integral in the above equation contains two quadratic phase terms and two linear phase terms. The quadratic phase terms represent deramping operations in azimuth and range, respectively, and the linear phase terms combined with the integral represent Fourier transforms in azimuth and range, respectively. Define $\mathcal{F}_{t\to f}(\cdot)$ as the Fourier transform, and define azimuth and range SPECAN operations as

$$\begin{aligned}
\mathrm{SPECAN}_a(\cdot)&=e^{j\pi K_a\eta^2}\mathcal{F}_{\eta'\to K_a\eta}\left((\cdot)e^{j\pi K_a\eta'^2}\right)\\
\mathrm{SPECAN}_r(\cdot)&=e^{-j\pi K_r't^2+j2\pi f_ct}\mathcal{F}_{\eta'\to K_r't-f}\left((\cdot)e^{-j\pi K_r't'^2}\right)dt'.
\end{aligned} \tag{14}$$

It holds for $\tilde{r}_l(\eta',t')=a_l\tilde{r}_{l,a}(\eta')\tilde{r}_{l,r}(t')$ that

$$\begin{aligned}
\mathrm{SPECAN_a}(\tilde{r}_{l,a}(\eta'))&=e^{j\pi K_a(\eta-\eta_l)^2}B_a\mathrm{sinc}(B_a(\eta-\eta_l))\\
\mathrm{SPECAN_r}(\tilde{r}_{l,r}(t'))&=e^{j\pi K_r'(t-t_l)^2+j2\pi f_c(t-t_l)}\\
&\qquad\times B_r'\mathrm{sinc}(B_r'(t-t_l)).
\end{aligned} \tag{15}$$

Therefore, we can replace the original matched filtering in (9) for focusing the RFI artifact $\tilde{r}(\eta',t')=\sum_{l=1}^{L}\tilde{r}_l(\eta',t')$ with the following equation:

$$\tilde{s}(\eta,t)_{\mathrm{SPECAN}}=\mathrm{SPECAN_r}(\mathrm{SPECAN_a}(\tilde{s}(\eta',t'))). \tag{16}$$

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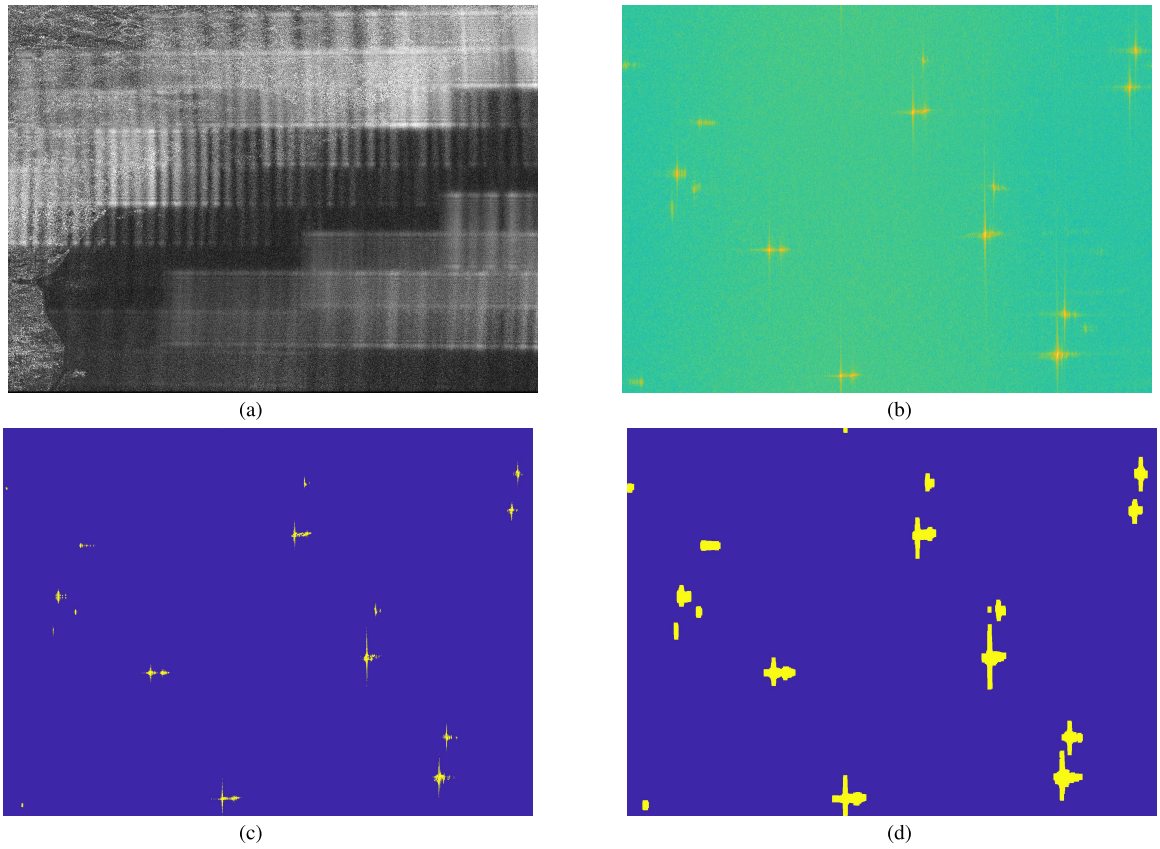

Fig. 2. Illustrative example for RFI focusing and removal. (a) Test image containing RFI. (b) RFI spectral sets Ω_{th} estimated using a threshold 20 dB higher than the mean spectrum amplitude (spectrum obtained via 2-D SPECAN). (c) RFI spectral set Ω_{aug} estimated after augmenting Ω_{th}. (d) Estimated RFI spectral set Ω_{aug}. In (b) and (c), the yellow pixels indicate the estimated spectral sets. The percentage of estimated RFI spectral points in (c) is 0.29% (i.e., the ratio between the number of spectral samples in the estimated RFI spectral set and the total sample number of the spectrum), and that in (d) is 2.0%.

Using the above operation, each individual RFI artifact $\tilde{r}_l(\eta', t')$ contained in $\tilde{s}(\eta', t')$ will be focused as a 2-D sinc response

$$Ca_l e^{j\phi(\eta,t;\eta_l,t_l)}\text{sinc}(B_a(\eta-\eta_l))\text{sinc}\big(B_r'(t-t_l)\big). \tag{17}$$

For simplicity, we define $\mathcal{F}_{(\eta,t)\to(f_a,f_r)}$ as the 2-D Fourier transform and rewrite (16) using an equivalent expression

$$\begin{aligned}\tilde{s}_{\text{SPECAN}_{2\text{D}}}(\eta,t) &= \underbrace{e^{j\pi K_a\eta^2 - j\pi K_r' t^2 + j2\pi f_c t}}_{\text{step 3: reramping}} \\ &\quad\times \underbrace{\mathcal{F}_{(\eta',t')\to(K_a\eta,-K_r't+f_c)}}_{\text{step 2: 2-D Fourier transform}}\left(\tilde{s}(\eta',t')\underbrace{e^{j\pi K_a\eta'^2 - j\pi K_r' t'^2}}_{\text{step1: deramping}}\right) \\ &= \text{SPECAN}_{2\text{D}}(\tilde{s}(\eta',t')).\end{aligned} \tag{18}$$

In this equation, the 2-D SPECAN consists of three steps: 1) time-domain multiplication by a deramping function; 2) perform 2-D Fourier transform; and 3) frequency-domain multiplication by a reramping function. Therefore, compared with the original matched filtering operation, the SPECAN method reduces the number of Fourier transforms to one and, therefore, improves computational efficiency. For the convenience of discussion, in this article, we will refer to the RFI-focused domain as a 2-D spectral domain since a 2-D Fourier transform is applied in the 2-D SPECAN processing.

In summary, we propose a 2-D SPECAN method to efficiently focus LFM RFI artifacts in the SLC image domain as point-like sinc responses in the 2-D spectral domain. An illustrative example of this procedure is shown in Fig. 2(a) and (b). This processing is similar to the SAR focusing concept that various algorithms are used for focusing a collection of SAR raw echoes backscattered from a point target as a sinc response [34]–[40]. The difference between the proposed 2-D SPECAN and these SAR focusing algorithms is that SAR focusing algorithms process SAR raw data to form SAR images, whereas our method processes SLC images with the goal of reducing the coupling between RFI and clean SAR images. With the reduction of this coupling via the proposed 2-D SPECAN, an RFI filter can be easily designed as discussed in the sequel.

IV. 2-D Spectral Notch Filtering for Removing Focused RFI Responses

Based on the method in Section III, we can remove pulsed RFI using the following generic processing steps.

1) Perform 2-D SPECAN to focus RFI from the SLC image domain to the 2-D spectral domain.
2) Remove the focused pulsed RFI in the 2-D spectral domain.
3) Perform inverse 2-D SPECAN to transform back into the image domain.

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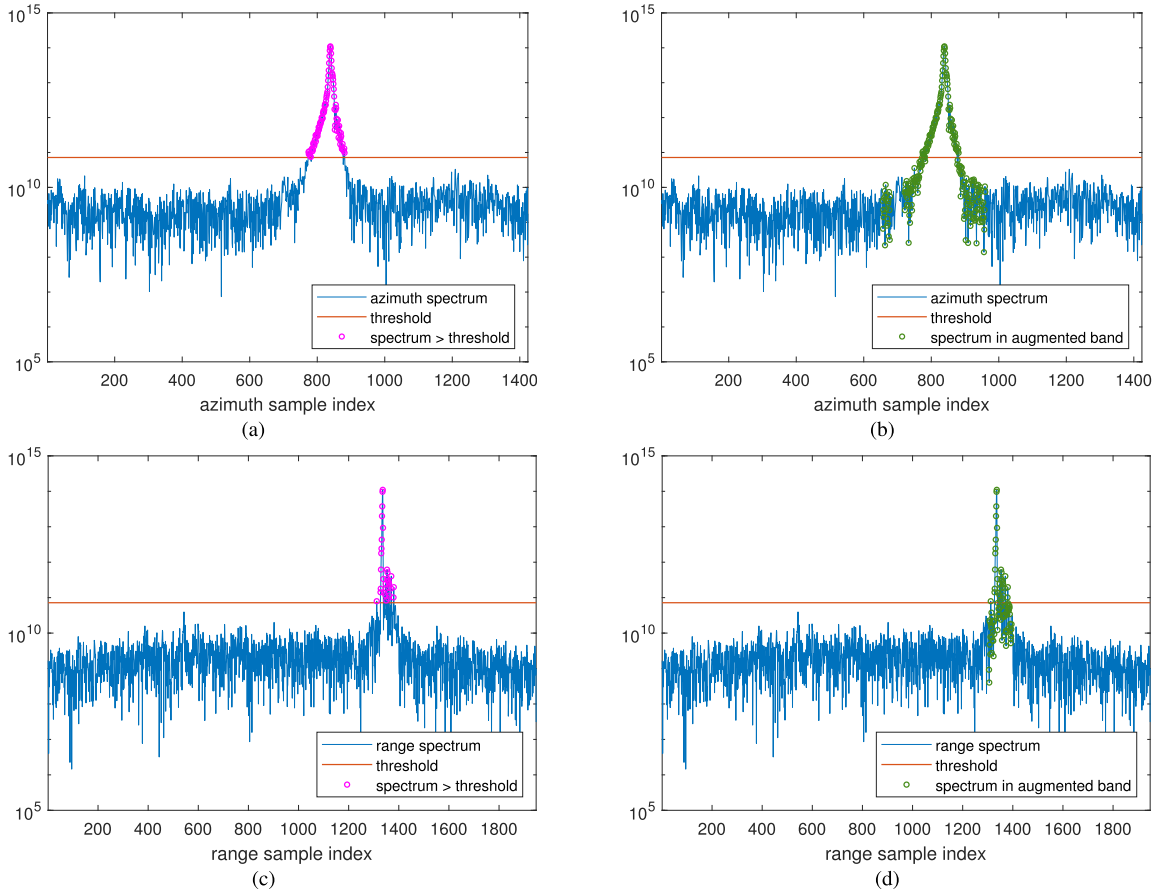

Fig. 3. Azimuth and range profiles of estimated RFI spectral set before and after augmentation. (a) and (c) Before spectral set argumentation, where the circs indicate the spectrum within the estimated RFI spectral set, i.e., $\Omega_{\text{th}} = \{(\eta, t) \mid \tilde{s}_{\text{SPECAN}_{2\text{D}}}(\eta, t) > \lambda\}$. (b) and (d) After spectral set augmentation, where the circs indicate the spectrum falls into the augmented RFI spectral set, i.e., Ω_{aug}. The threshold is selected to be 20 dB above the mean spectrum amplitude. The azimuth and range spectrum profiles plotted using blue lines correspond to the 1336th vertical line and 839th horizontal line of the spectrum in Fig. 2(b), respectively (a) Azimuth profile with thresholding. (b) Azimuth profile with thresholding and band augmentation. (c) Range profile with thresholding. (d) Range profile with thresholding and band augmentation.

In the first step, a spatially extended LFM RFI artifact is focused as a point-like response, and its energy mainly lies in the main lobe of its sinc shape, $\eta \in [-1/2B_a, 1/2B_a], t \in [-1/2B_r', 1/2B_r']$. On the contrary, the underlying clean SAR signal and noise are not focused. They are flat signals after transforming into the 2-D spectral domain by the 2-D Fourier transform. The removal of RFI is then can be viewed as a problem of removing isolated points (i.e., the focused RFI response) in a flat background signal (i.e., the underlying clean image signal and noise). Therefore, it is important to choose a proper method for the removal of focused RFI responses in the 2-D spectral domain. In this article, we use 2-D notch filtering in the 2-D spectral domain for this task.

A. Notch Filtering for Removing RFI Responses

Notch filtering removes unwanted components of a signal by constructing notches in the signal spectrum, which discards the spectrum within the frequency band of the unwanted signal. It is very simple and efficient for suppressing narrowband RFI. This technique is usually used as a 1-D filter. In this article, we extend the notch filtering approach as a 2-D filter for RFI removal, which works by constructing notches in the 2-D spectral domain. The constructed notches are designed to cover RFI point responses such that the RFI contribution will be filtered out.

Constructing spectral notches requires the knowledge of the positions of the focused RFI point responses in the 2-D spectral domain, which, however, cannot be known *a priori* and, therefore, must be estimated from data. In this article, we extract the required spectral positions of the RFI point responses via two steps.

1) Select a threshold λ, and extract the position set of the focused RFI by taking the spectral position set of 2-D spectra with amplitudes exceeding this threshold, i.e.,
$$\Omega_{\text{th}} = \left\{(\eta, t) \mid |\tilde{s}_{\text{SPECAN}_{2\text{D}}}(\eta, t)| > \lambda\right\} \tag{19}$$
where Ω_{th} is the spectral position set of the focused RFI responses. λ should be chosen α dB above the background signal spectrum. In this article, we use $\alpha = 20$ as an empirical setting.
2) Augment the spectral position set by including the neighborhood of each position in the initial set Ω_{th}
$$\Omega_{\text{aug}} = \{(\eta, t) \mid (\eta, t) \in \mathcal{N}_{\Delta_a, \Delta_b}(a, b), (a, b) \in \Omega_{\text{th}}\} \tag{20}$$

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where $\mathcal{N}_{\Delta_a,\Delta_b}(a,b) = \{(\eta,t) \mid \eta \in [a-\Delta_a/2, a+\Delta_a/2], t \in [b-\Delta_b/2, b+\Delta_b/2]\}$ is the neighborhood of position (a,b) with width Δ_a and Δ_b in azimuth and range, respectively.

In the above processing, the first step obtains an initial position set estimation exploiting the fact that the focused RFI point responses have a significantly larger amplitude than the flat background spectrum that corresponds to the underlying clean SAR signal and noise. The threshold λ should be chosen much larger than the amplitude level of the background spectrum to avoid mistakenly taking clean positions as RFI positions. On the other hand, a large threshold usually only extracts partial support of the main lobe of the focused sinc-shaped RFI response, making partial positions not included in the estimated spectral position set [as the example shown in Fig. 3(a)]. The second step is used to fix this problem, to include more RFI spectral positions by taking the neighborhood of each extracted RFI spectral position into account.

Using the estimated RFI spectral set, we define a 2-D mask accounting for the 2-D notch filtering in the spectral domain

$$w(\eta,t) = \begin{cases} 0, & (\eta,t) \in \Omega_{\text{aug}}; \\ 1, & (\eta,t) \notin \Omega_{\text{aug}}. \end{cases} \tag{21}$$

Multiplying this mask with the SPECAN output $\tilde{s}_{\text{SPECAN}_{2D}}(\eta,t)$ is equivalent to notching out the spectrum within the estimated RFI spectral set Ω_{aug}. Using this notation, the filtered spectrum can be expressed as

$$\tilde{s}_{\text{notch}}(\eta,t) = w(\eta,t)\tilde{s}_{\text{SPECAN}_{2D}}(\eta,t). \tag{22}$$

B. Inverse 2-D SPECAN

After applying 2-D notch filtering to RFI-focused data in the spectral domain, we perform the inverse operation of the 2-D SPECAN to transform $\tilde{s}_{\text{notch}}(\eta,t)$ back into the original image domain. This inverse operation can be derived from (18) and is provided as follows:

$$\begin{aligned} &\hat{s}(\eta',t') \\ &= e^{-j\pi K_a \eta'^2 + j\pi K_r' t'^2} \\ &\quad \mathcal{F}^{-1}_{(\eta',t')\to(K_a\eta,-K_r't+f_c)}\left(\tilde{s}_{\text{notch}}(\eta,t)e^{-j\pi K_a\eta^2+j\pi K_r't^2-j2\pi f_c t}\right) \\ &= \text{SPECAN}^{-1}_{2D}(\tilde{s}_{\text{notch}}(\eta,t)). \end{aligned} \tag{23}$$

After this processing, most of the RFI contribution will be removed, and a clean SLC image can be obtained. A comparison of RFI removal results obtained with and without spectral set augmentation is provided in Fig. 4.

C. Analysis of Signal Loss

It should be pointed out that the spectral point set Ω_{aug} also contains a partial clean image spectrum, which will be discarded along with the focused RFI responses by the filtering operation. This will result in a signal loss in the final processed image. With the assumption of flat signal spectrum, the expectation of the signal loss for the clean image $\tilde{x}(\eta',t')$ can be characterized by the ratio between the numbers of spectral samples detected to contain RFI (denoted as N_{RFI}) and the total spectral sample numbers (denoted as $N_a N_r$), i.e.,

$$\rho = \frac{N_{\text{RFI}}}{N_a N_r}. \tag{24}$$

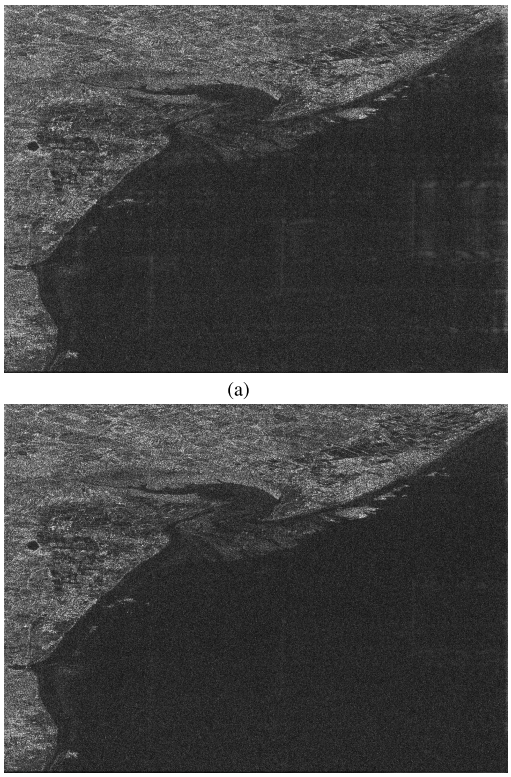

Fig. 4. Comparison of RFI removal results obtained with and without spectral set augmentation. (a) Processed without spectral set augmentation. (b) Processed with spectral set augmentation.

Intuitively, N_{RFI} is the number of yellow pixels in Fig. 2(d), and $N_a N_r$ is the total pixel number in the image. If we ignore weak RFI sidelobes outside the spectral point set Ω_{aug}, the above ratio of signal loss is identical to the relative mean square error of the underlying clean image plus noise. In Sentinel-1 imagery, this ratio is typically smaller than 5%. It is clear that this signal loss is very small.

V. Processing and Implementation Details

Based on the above 2-D SPECAN-based RFI removal method, in this section, we provide a simplified processing flow and discuss some implementation details.

A. Simplified Processing

In order to reduce the computation cost for the 2-D SPECAN-based RFI removal method, here, a simplified processing flow is designed. The reduction in computation cost is obtained by removing the exponential phase multiplications in the third step of the 2-D SPECAN and the first step of the inverse SPECAN, respectively. To show this relation, we express the overall processing flow of 2-D SPECAN, notch filtering, and inverse 2-D SPECAN as follows:

$$\begin{aligned} &\hat{s}(\eta',t') \\ &= e^{-j\pi K_a\eta'^2+j\pi K_r't'^2}\mathcal{F}^{-1}_{(\eta',t')\to(K_a\eta,-K_r't+f_c)} \\ &\quad \left(e^{-j\pi K_a\eta^2+j\pi K_r't^2-j2\pi f_c t}w(\eta,t)e^{j\pi K_a\eta^2-j\pi K_r't^2+j2\pi f_c t}\right. \\ &\quad \left.\mathcal{F}_{(\eta',t')\to(K_a\eta,-K_r't+f_c)}\left(s(\eta',t')e^{j\pi K_a\eta'^2-j\pi K_r't'^2}\right)\right). \end{aligned} \tag{25}$$

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In this equation, the third line equals $w(\eta, t)$ because the two exponential phase terms in this line are conjugate to each other. Thus, we can simplify the processing steps as

$$\begin{aligned}\hat{s}(\eta', t') &= e^{-j\pi K_a \eta'^2 + j\pi K_r' t'^2} \mathcal{F}^{-1}_{(\eta', t') \to (K_a \eta, -K_r' t + f_c)} \\ &\times \left(w(\eta, t) \mathcal{F}_{(\eta', t') \to (K_a \eta, -K_r' t + f_c)} \left(s(\eta', t') e^{j\pi K_a \eta'^2 - j\pi K_r' t'^2} \right) \right)\end{aligned} \tag{26}$$

which, compared with the original processing steps, saves two exponential phase multiplications.

In this equation, $w(\eta, t)$ is obtained from the spectrum generated using the Fourier transform $\mathcal{F}_{(\eta', t') \to (f_a, f_r)}$ followed by variable mapping $(f_a, f_r) \to (K_a \eta, -K_r' t + f_c)$, thresholding, and spectral position set augmentation. The variable mapping is helpful for exactly relating the Fourier transformed data to the reference plane (η, t), but this relation is actually redundant for the notch filtering step, and removing this variable mapping is helpful for further simplifying the processing step. To obtain this simplification, we define $\tilde{w}(f_a, f_r) = w(f_a/K_a, (f_c - f_r)/K_r')$ and reexpress (26) as

$$\begin{aligned}\hat{s}(\eta', t') &= e^{-j\pi K_a \eta'^2 + j\pi K_r' t'^2} \mathcal{F}^{-1}_{(\eta', t') \to (f_a, f_r)} \\ &\times \left(\tilde{w}(f_a, f_r) \mathcal{F}_{(\eta', t') \to (f_a, f_r)} \left(s(\eta', t') e^{j\pi K_a \eta'^2 - j\pi K_r' t'^2} \right) \right).\end{aligned} \tag{27}$$

The spectral set estimation method introduced previously also works in the (f_a, f_r) domain since (f_a, f_r) and (η, t) essentially represent the same domain where RFI artifacts are focused as point-like responses. Therefore, $\tilde{w}(f_a, f_r)$ can be obtained using the same method introduced in Section IV. It is also worth noting that the simplified processing steps also avoid requiring the prior knowledge of parameter f_c. However, the FM rate parameters, K_a and K_r', still need to be provided, as will be discussed in the following.

B. Discrete Processing

In the above discussions, continuous signals are used to introduce the theoretical basics of this study for the simplicity of expressing signals with analytical forms. However, as SAR SLC images are discrete samples, the processing flow in (27) should be translated into its discrete form. To this end, we express the original SLC image as the samples of $s(\eta', t')$, i.e.,

$$s[p, q] = s(p\delta_a, q\delta_r) \tag{28}$$

where p and q are azimuth and range pixel indexes, respectively, δ_a and δ_r are the azimuth and range pixel spacings, respectively. Similarly, we define the output image of the proposed filtering method as the samples of $\tilde{s}(\eta', t')$, i.e., $\tilde{s}[p, q] = \tilde{s}(p\delta_a, q\delta_r)$.

The Fourier transform in (27) needs to be replaced with fast Fourier transform (FFT). The fftshift operation commonly used to shift the zero frequency to the center of the FFT spectrum is not necessary here, as the 2-D RFI spectral set estimation and notch filtering can still work with the original FFT spectrum directly.

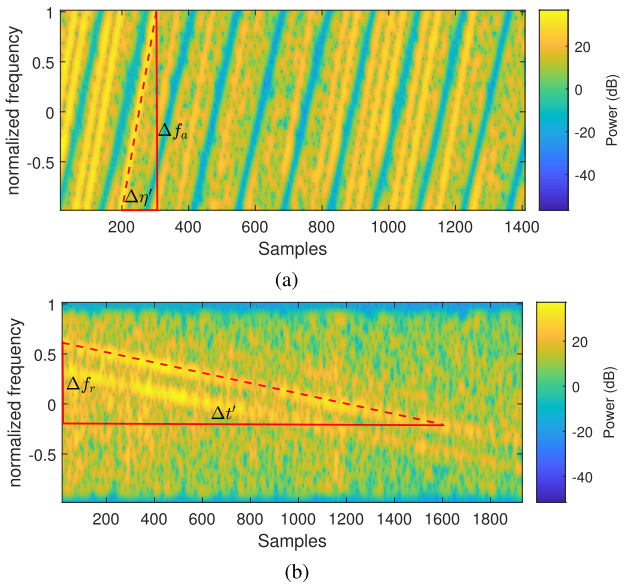

Fig. 5. Estimating azimuth and range FM rate from (a) azimuth and (b) range spectrograms of the SLC image. The estimated azimuth and range FM rates, i.e., the slope of the dash lines in the subfigures, are $\hat{K}_a = 2.268 \times 10^3$ Hz/s and $\hat{K}_r = -1.109 \times 10^{12}$ Hz/s.

The discrete operation of spectral point set augmentation can be performed via 2-D convolution. Specifically, define $\tilde{w}_{\text{th}}[n, m]$ as mask matrix relating the initial spectral set of focused RFI responses, where $\tilde{w}_{\text{th}}[n, m] = 0$ if the $[n, m]$th spectrum sample exceeds the predefined threshold λ, and otherwise, $\tilde{w}_{\text{th}}[n, m] = 1$. Define a convolution kernel $e[n, m] = 1,\ n = 1, \ldots, 2L_a + 1, m = 1, \ldots, 2L_r + 1$, where L_a and L_r are the number of left or right neighboring points to be included in the augmented set. Using this notation, the augmented set can be expressed as

$$\begin{aligned}\tilde{w}_{\text{aug}}[n, m] &= \tilde{w}_{\text{th}}[n, m] * e[n, m] \\ \tilde{w}_{\text{aug}}[n, m] &= 1 \quad \forall \tilde{w}_{\text{aug}}[n, m] > 1\end{aligned} \tag{29}$$

where $*$ denote 2-D convolution.

Finally, denote the discrete azimuth and range deramping function as $d_a[p] = e^{j\pi \hat{K}_a (p\delta_a)^2}$, $d_r[q] = e^{-j\pi \hat{K}_r' (q\delta_r)^2}$, where $\hat{K}_a$ and $\hat{K}_r$ are the azimuth and range FM rate estimates (the estimation will be discussed in the sequel), and denote FFT as $\mathcal{F}_{(p,q) \to (n,m)}$. With these notations, the 2-D SPECAN-based discrete processing can be expressed as

$$\begin{aligned}\hat{s}[p, q] &= d_a^*[p] d_r^*[q] \mathcal{F}^{-1}_{(p,q) \to (n,m)} \\ &\times \left(\tilde{w}_{\text{aug}}(n, m) \mathcal{F}_{(p,q) \to (n,m)} \left(s[p, q] d_a[p] d_r[q] \right) \right)\end{aligned} \tag{30}$$

which is a processing pipeline of elementwise product, 2-D FFT, elementwise product, 2-D IFFT, and elementwise product.

C. FM Rates' Estimation

The proposed processing method requires the knowledge of azimuth and range FM rates, K_a and K_r', to determine the

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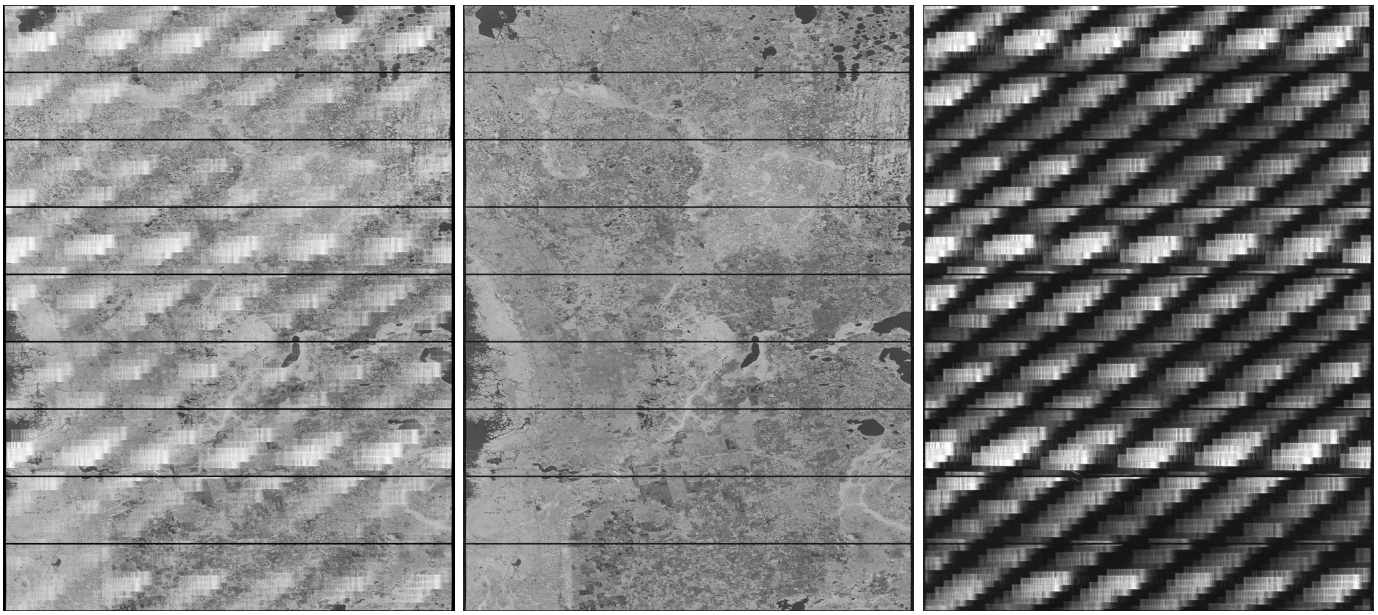

(a) (b) (c)

Fig. 6. Experiment #1 using a Sentinel-1 VH image acquired over the land region in Florida. The experiment image is the VH channel, IW1 of S1A_IW_SLC__1SDV_20161208T232833_20161208T232900_014295_01723 1_7ABD.zip. This image is polluted by pulse LFM RFI with a long duration in azimuth, which is assumed to be caused by another C-band SAR. The image size is 13 446 × 21 218. The range is horizontal, and azimuth is vertical. (a) Input. (b) Output. (c) Removed.

exponential phase function $e^{j\pi K_a \eta'^2 - j\pi K_r' t'^2}$ in (27) (i.e., the deramping function). In this study, we estimate the two FM rate parameters from the azimuth and range spectrograms of the SLC image. This estimation procedure for K_a and K_r' can be done by calculating the slope of the RFI-induced bright line in the spectrogram, using the following equation:

$$\hat{K}_a = \frac{\Delta f_a}{\Delta \eta'} \quad \hat{K}_r' = \frac{\Delta f_r}{\Delta t'} \tag{31}$$

where Δf_a and Δf_r are the azimuth and range frequency spans of the RFI spectrogram line, respectively, and $\Delta \eta'$ and $\Delta t'$ are their associated time spans. This method requires that the RFI line should be clearly visible in the spectrogram. Since RFI often has significantly higher signal level than the clean SAR signal plus noise, this requirement is easy to satisfy. Fig. 5 shows an example of estimating the FM rates from azimuth and range spectrograms. In this example, the estimated azimuth and range FM rates using (31) are $\hat{K}_a = 2.268 \times 10^3$ and $\hat{K}_r' = -1.109 \times 10^{12}$, respectively, and they are used to obtain the illustrative results in Figs. 2, 3, and 12.

D. Blockwise Processing Strategy

SAR data and images usually have very large sizes, which exceeds the processing capacity of computing platforms. Blockwise processing is a useful strategy to solve this problem, and it is used in various SAR data processing problems, e.g., image focusing, autofocusing [41], coregistration, and SAR deep learning. In our study, we incorporate blockwise processing with the proposed RFI filtering method, leading to the following overall processing flow.

1) Segment the SLC image to be processed into blocks with predefined block size $W_a \times W_r$.
2) For each block, apply the proposed method to remove RFI, using the processing pipeline of deramping, 2-D FFT, spectral notch filtering, inverse FFT, and reramping. At this step, multithreading and parallel processing can be adopted.
3) *Block Recombination:* This step can be omitted if each filtered data block is written into a memory device using proper addresses or just overrides the original data block in the memory device.

VI. Experiments

In this section, we present five experiments with Sentinel-1 SLC images acquired in the interferometric wide (IW) swath mode to demonstrate the powerful ability of the proposed method in RFI removal. The images used in our experiments are polluted by real pulse LFM RFI. IW is the default data acquisition mode of Sentinel-1 with dual polarization (VV and VH), and it has three subswath called IW1, IW2, and IW3. In experiments #1 and #2, we use a pair of dual-polarization IW1 images over the land region. In experiments #3 and #4, we use a pair of dual-polarization IW1 images containing sea area. In experiment #5, a small area of a VH IW2 image is studied.

Sentinel-1 IW images consist of multiple bursts with a burst size of around 1500 pixels in azimuth and larger than 20 000 pixels in range. To make the proposed algorithm applicable to such a large size, we use blockwise processing with one burst length (its exact value depends on the burst to be processed) in azimuth and 2000 (fixed for all bursts) in range. Results of processed SLC images $\hat{s}[p, q]$ are presented as gray-scale JPG images. The colorbar of all used Sentinel-1 images are the same, which maps amplitude values from [0,80]

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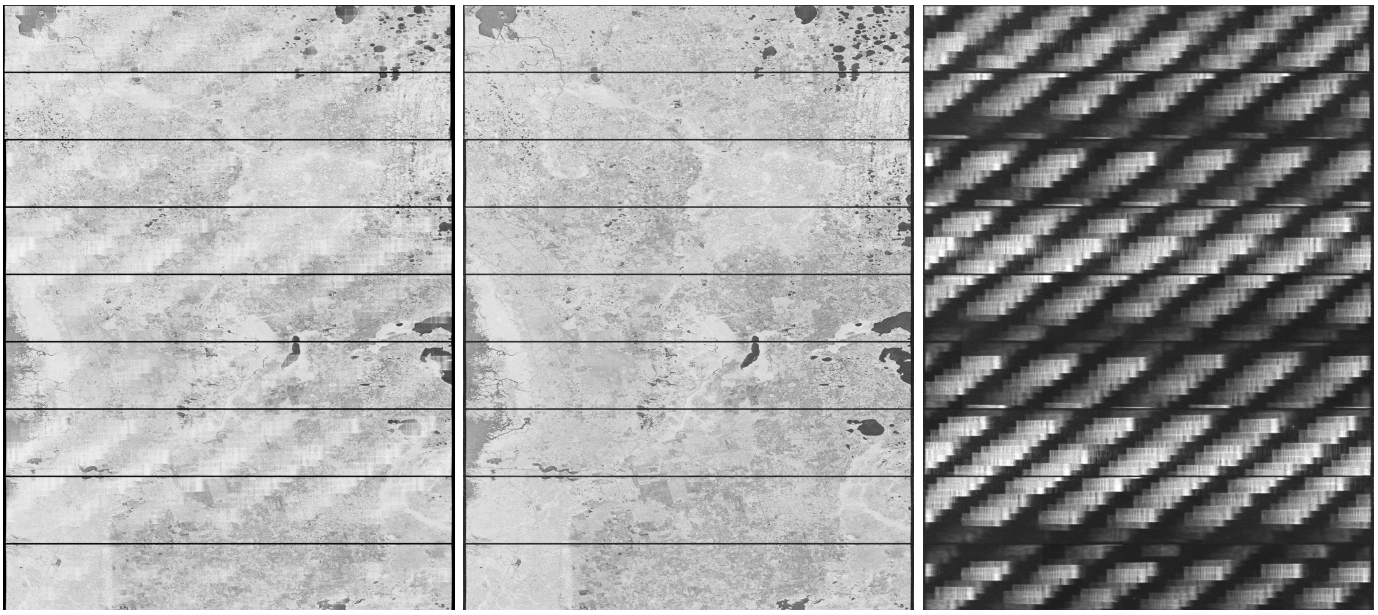
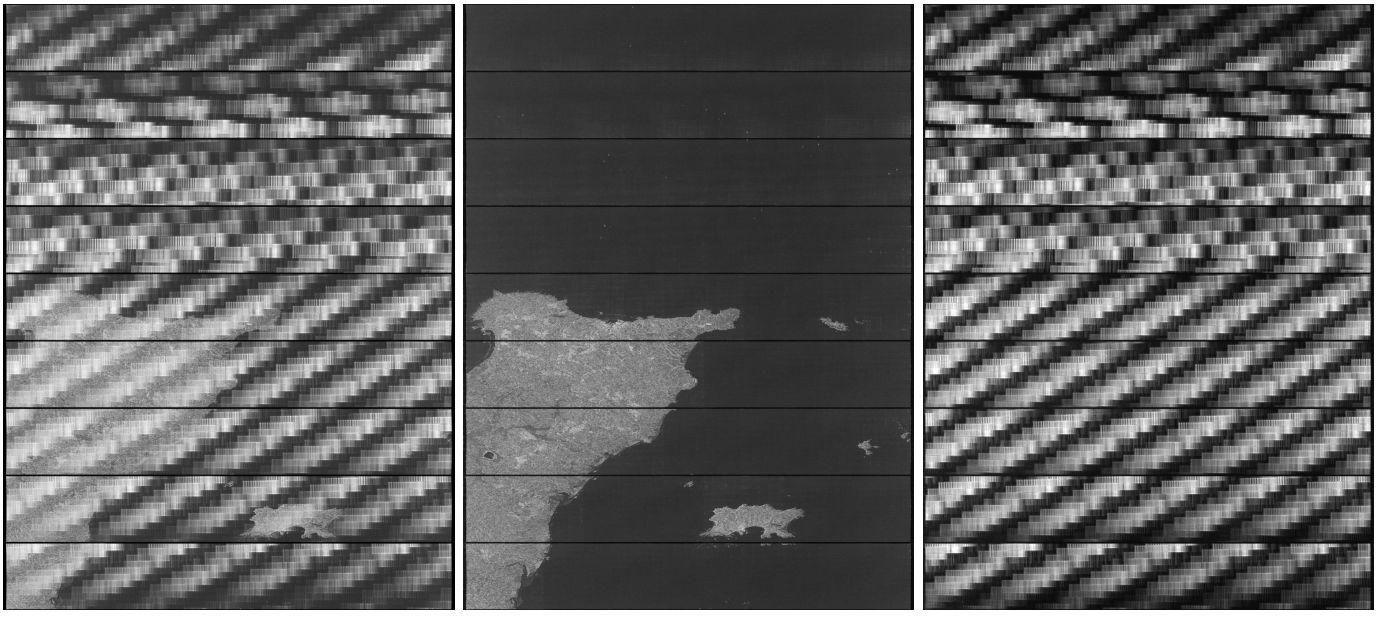


(a) (b) (c)

Fig. 7. Experiment #2 using a Sentinel-1VV image acquired over land region in Florida. The experiment image is the VV channel of the same Sentinel-1 archived data in the previous experiment. (a) Input. (b) Output. (c) Removed.



(a) (b) (c)

Fig. 8. Experiment #3 using a Sentinel-1 VH image acquired over sea region in North France. The experiment image is the VH channel, IW1 of S1B_IW_SLC__1SDV_20190411T0622 53_20190411T062320_015755_01D915_2A6A.zip. This image is also polluted by long-duration pulsed LFM RFI assumed to be caused by another C-band SAR. The image size is $13563 \times 23\,055$. (a) Input. (b) Output. (c) Removed.

to [0,255] as gray-scale images. For clear comparison, the gray-scale images of the original SLC images $\tilde{s}[p, q]$ and removed RFI components (obtained via $\hat{s}[p, q] - \tilde{s}[p, q]$) are also provided along with the gray-scale images of the processed SLC images.

At the end of this section, we will perform experiments using simulated data to quantitatively compare the 2-D SPAECAN method with PCA and RPCA methods. Experiment details and result analyses are discussed in the following.

A. Experiment #1

In the first experiment, the used image is the VH channel of the IW1 in the Sentinel-1 data S1A_IW_SLC__1SDV_20161208T232833_20161208T232900_014295_017231_7ABD.zip. The acquisition location of this data is around Florida. These data are polluted by pulsed LFM RFI with a very long duration in azimuth, which causes a series of bright strips in the SLC image, as shown in Fig. 6. The source of the RFI is assumed to be another C-band SAR as reported in [2].

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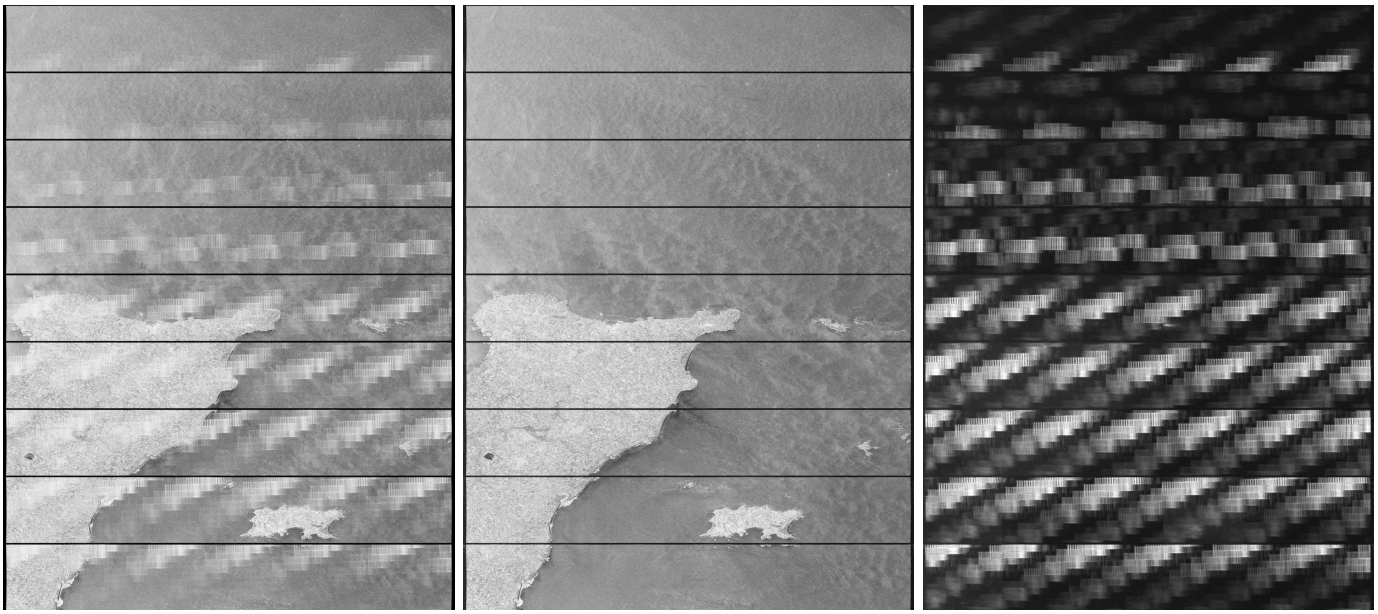
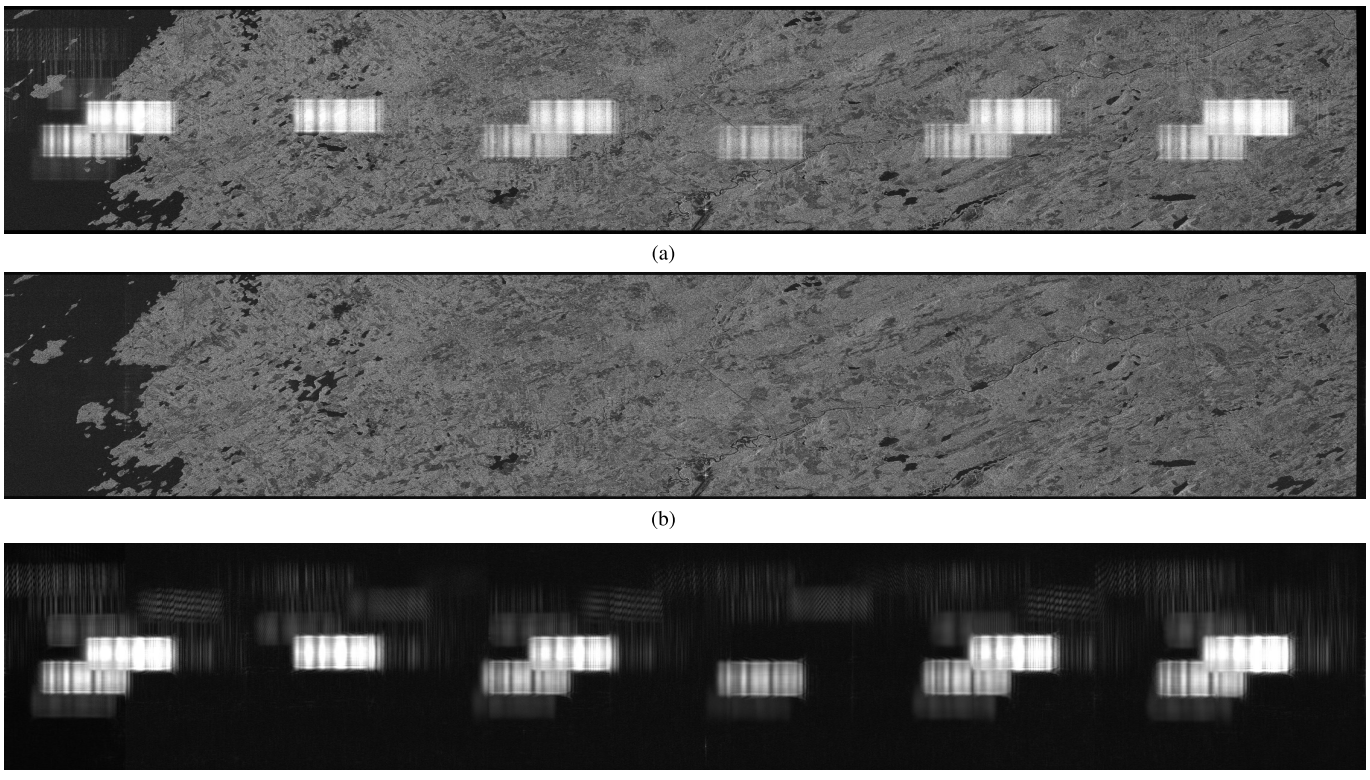


(a) (b) (c)

Fig. 9. Experiment #4 using a Sentinel-1 VV image acquired over sea region in North France. The experiment image is the VV channel of the same Sentinel-1 archived data used in the previous experiment. (a) Input. (b) Output. (c) Removed.



(a)

(b)

(c)

Fig. 10. Experiment #5 using a Sentinel-1 VH image acquired over north Sweden near the coast of Baltic sea. The experiment image is the burst 4, VH channel, IW2 of S1A_IW_SLC__1SDV_20190607T050519_20190607 T050547_027569_031C74_FD1D.zip. The image size of this burst is $1519 \times 26\,756$. (a) Input. (b) Output. (c) Removed.

The processed image and the associated removed RFI components are visualized as gray-scale images in Fig. 6(b) and (c), respectively. The results show that some regions buried by strong RFI artifacts now are clean in the processed image. In the gray-scale image of the removed RFI components, there do not exist significant land reflectivity signatures, showing that the proposed method effectively preserves the underlying reflectivity information.

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B. Experiment #2

In this experiment, the VV channel of the IW1 in the same Sentinel-1 data in the previous experiment is used. The gray-scale image of the VV SLC image is shown in Fig. 7(a). The intensity level of the VV channel is higher than the VH channel, and it can be seen that the signal-to-interference ratio (SIR) is also higher than that in the previous experiment. High SIR will decrease the performance of the PCA method because this method requires a high interference-to-signal ratio to estimate the RFI component [5]. The proposed method does not have this drawback, as the clean processed image shown in Fig. 7(b). The gray-scale image of the removed RFI component is shown in Fig. 7(c). Comparing Figs. 6(c) and 7(c), we can find that the power levels of the RFI artifacts observed in the VV and VH channels are close. The reason for this behavior is that the RFI source might be multiple-polarization SARs. Another detail that can be observed is that the strips (one strip corresponds to one pulsed LFM RFI) of the RFI artifacts do not have the same position in VV and VH images. This might be due to the difference in the timing of transmitting and receiving pulses of different polarisations.

C. Experiment #3

The above two experiments consider a case of the land region. In this experiment, we consider a case containing sea area. The tested image in this experiment is the VH channel of the IW1 in the Sentinel-1 data S1B_IW_SLC__1SDV_20190411T0622 53_20190411T062320_015755_01D915_2A6A.zip. Its gray-scale image is shown in Fig. 8(a), where the region is near north France. Similar to the previous experimental images, this image is also polluted by pulsed LFM RFI, and the source is assumed to be another C-band SAR, as reported in [2].

The gray scales of the processed image and the removed RFI components are shown in Fig. 8(b) and (c), respectively. Since VH reflectivity over sea area is usually weak, we can use this property to investigate residual RFI artifacts in the processed image. As shown in the processed image, there still exist some weak residual RFI artifacts in the processed image. The residue can be reduced if we use a lower threshold for estimating Ω_{th} and a larger neighborhood Δ_a and Δ_b at the cost of increasing the signal loss of the underlying clean image.

D. Experiment #4

In this experiment, we use the VV channel of the IW1 in the same Sentinel-1 data mentioned in experiment #3. The original image, its processed version, and the removed RFI components are visualized as gray-scale images, which are shown in Fig. 9(a)–(c), respectively. In the processed image, we can see that there are no significant residual RFI artifacts, showing that the residue level is much lower than the signal level of the clean VV image. It is notable that such RFI images can be used to coarsely localize RFI sources by combing a pair of ascending and descending acquisitions [42].

E. Experiment #5

In the last experiment, we consider pulsed LFM RFI caused by ground radar. The used image is the fourth burst of IW2, VH channel, in the Sentinel-1 data S1A_IW_SLC__1SDV_20190607T050519_20190607T050547_027569_031C74_FD1D.zip. In this Sentinel-1 data, only this burst contains visible RFI artifacts, so we extract this burst only for experiment and the rest part of this Sentinel-1 data is not used. We find in Sentinel-1 archived data that there exist frequent occurrences of pulsed LFM RFI with roughly 3-MHz bandwidth over the acquisition region of this data. For example, S1A_IW_SLC__1SDV_20210517T160625_20210517T160652_037930_0479FC_06F3.zip also contains pulsed LFM RFI artifacts over the same

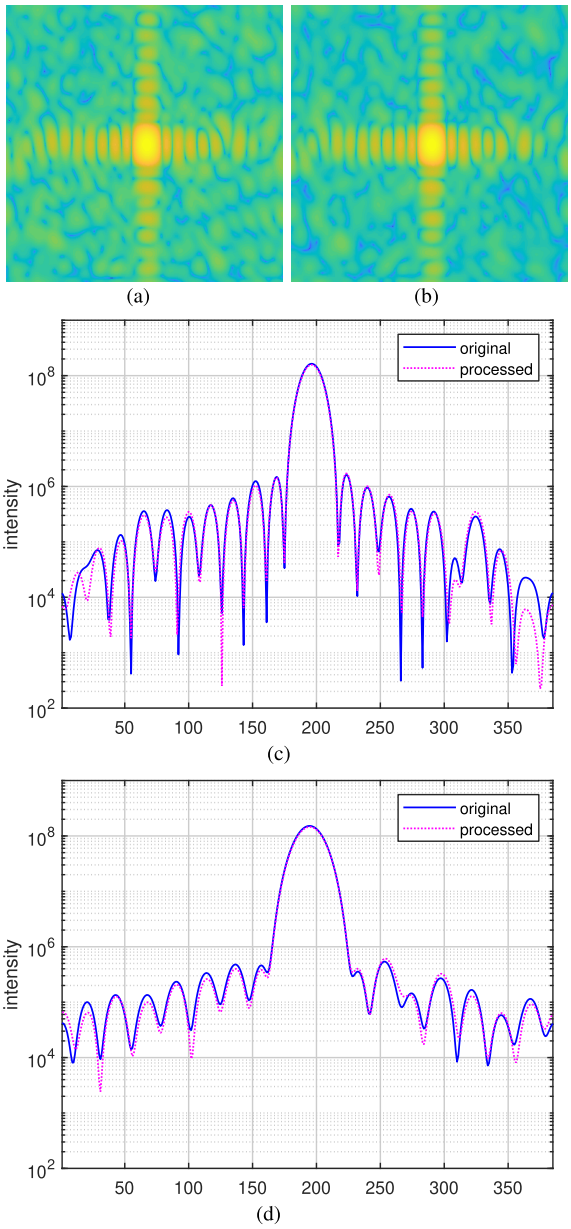

Fig. 11. Azimuth and range profiles of an interpolated point target before and after applying the proposed method. The point scatterer in this image is the [362, 3230]th pixel of burst 7 in experiment #1. The oversampling rate used for interpolation is 32. Peak intensities are $1.6386e+08$ before filtering, and $1.5668e+08$ after filtering. The reduction in peak intensity is 0.0438, and the signal loss ratio ρ of the point target is less than this value. (a) Original log intensity image. (b) After filtering. (c) Range profile. (d) Azimuth profile.

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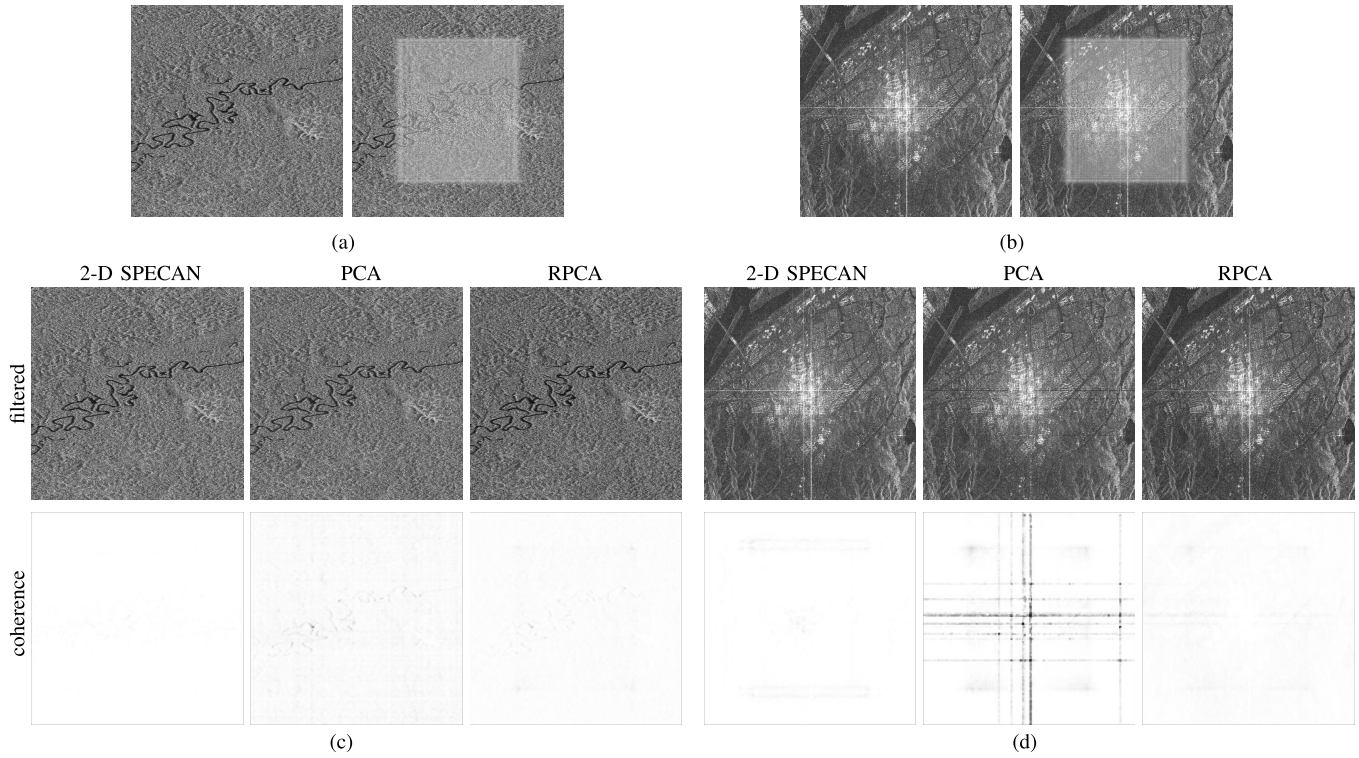

Fig. 12. Comparison between 2-D SPECAN, PCA, and RPCA. (a) and (b) Reference images and their RFI-polluted versions (using simulated pulsed LFM RFI with −3-dB SIR). (c) Results for (a). (d) Results for (b). The image size is 2000 × 2000. Colormap from the amplitude image to gray-scale image: $[0, \alpha] \rightarrow$ [0,255], where α is the 95% percentiles of the amplitude values of the reference images. Colormap from the coherence image to gray-scale image: $[0, 1] \rightarrow$ [0,255]. Black represents low coherence, and white represents high coherence.

region. Thus, we think that the associated RFI source is a ground radar working continuously, which might be a C-band weather radar or air surveillance radar. The original image, its processed version, and the removed RFI components are visualized as gray-scale images in Fig. 10(a)–(c), respectively. The results show that the proposed method outputs a very clean image while do not cause significant signal loss because the gray-scale image in Fig. 10 does not contain noticeable land reflectivity signatures.

F. Analysis of Signal Loss Using Strong Point Target

In our previous works [5], [43], we have designed PCA and block subspace filter (BSF) for RFI removal, where BSF is a block-wide processing filter with its kernel being equivalent to the PCA method in [5]. PCA has a significant drawback of losing strong point targets. To investigate whether the proposed 2-D SPECAN method has this drawback, we perform a quantitative analysis using a strong point target. Fig. 11 shows the intensity profile of the strong point target before and after applying the proposed method. The point target is taken from burst 7 in experiment #1. Since the point target has a very high-intensity level, its interference-to-signal ratio is so high that the RFI artifacts are negligible for this point. In this example, the peak intensity before applying the proposed method is 1.6386×10^8 and is reduced to 1.5668×10^8 after applying the proposed method. This reduction is due to the removal of both RFI and clean signals within the focused RFI point set Ωaug, so, in this example, the relative loss of the point target, ρ, is less than 0.0438. From this example, we can clearly see that the proposed SPECAN method preserves point targets well.

G. Comparison With PCA and RPCA

In this section, we use simulation-based experiments to compare the proposed 2-D SPECAN filter with PCA and RPCA for RFI removal [5]. In our simulations, two clean SLC images are taken as the references: one of which is a homogeneous scene of forest and the other is a heterogeneous scene of the urban region having strong point scatterers. An interference artifact of an LFM pulse is simulated via the omega-k algorithm, using the following parameters: center frequency of SAR 5.4 GHz, sampling frequency of 64.3 MHz, pulse repetition frequency of 1717 Hz, FM rate of SAR 1.07×10^{12} Hz/s, closest range of the scene center of 850 Km, platform velocity of 7100 m/s, FM rate of RFI -4.6×10^{11} Hz/s, and pulse duration of RFI of 13 μs. The artifact is then scaled according to different SIRs and added to the clean reference images to simulated RFI-polluted versions for experiments. For PCA, the first K principal components are filtered out to removed RFI, where $K = 10$ in our experiment.

The processing results under the setting of −3-dB SIR are shown in Fig. 12, where the first to fourth rows contain the reference images, filtered images, and the coherence between the reference and filtered images, respectively. Here, the coherence is used to measure the phase consistence between the

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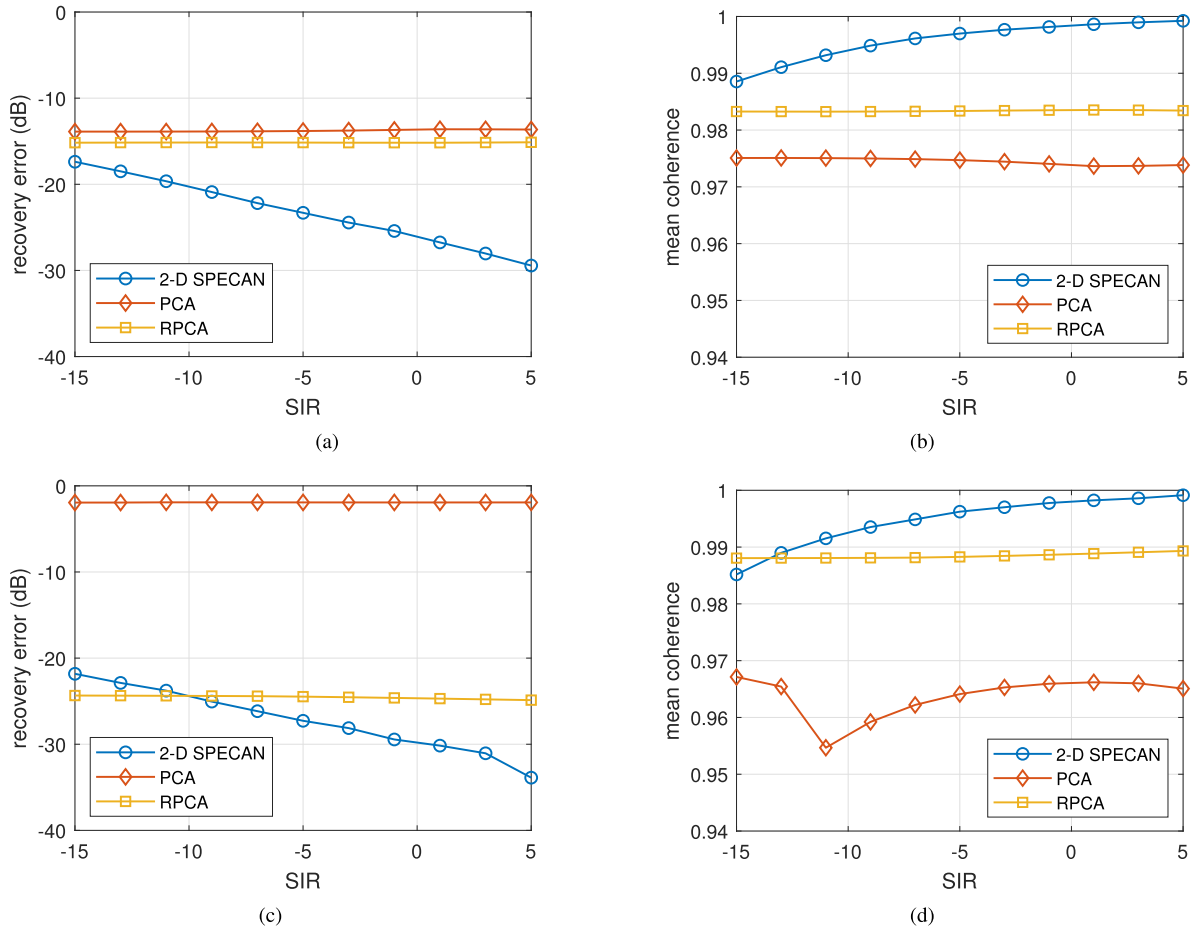

Fig. 13. Recovery error and mean coherence versus SIR. (a) and (b) are obtained with the homogeneous scene in Fig. 12(a). (c) and (d) are obtained with the heterogeneous scene in Fig. 12(b). The high recovery error of PCA in (c) is mainly due to the loss of strong point scatterers.

reference images and their filtered ones, which is defined as

$$\gamma_{i,j} = \frac{\left|\mathbb{E}\left\{\hat{S}_{i,j} S_{i,j}^{*}\right\}\right|}{\sqrt{\mathbb{E}\{|\hat{S}_{i,j}|^2\}\mathbb{E}\{|S_{i,j}|^2\}}} \tag{32}$$

where $S_{i,j}$ and $\hat{S}_{i,j}$ are the $[i, j]$th pixel of the original and filtered images, respectively. For numerical calculation, we replace the expectation operator, $\mathbb{E}$, with local averaging using a window of 11×11 pixels. From the filtered images in Fig. 12, we can find that: 1) all the three methods can effectively remove the LFM RFI artifact, especially in the case of homogeneous background and 2) PCA loses strong point scatterers in the case of heterogeneous scene, whereas 2-D SPECAN and RPCA do not.

To quantitatively compare the three methods, we perform a series of simulation-based experiments using different SIR settings and compute the (relative) recovery error defined as $10\log_{10}(||\hat{\mathbf{S}} - \mathbf{S}||_F^2/||\mathbf{S}||_F^2)$ and the mean coherence (the average of all $\gamma_{i,j}$). The recovery error and mean coherence obtained under SIR $= -15, -8, \ldots, 5$ are shown in Fig. 13 and partly listed in Table I. It is shown in the figure that all three methods can obtain high coherence (>0.94) with respect to the reference images, demonstrating that they are phase-preserving. In Fig. 13(d), the PCA curve shows a nonmonotonic trend with SIRs, which is due to the following reason: smaller SIRs can lead to higher accuracy via PCA and, therefore, higher coherence, whereas larger SIRs can cause worse recovery accuracy via PCA, but the coherence can still increase since the SIR increases. Among these methods, the 2-D SPECAN filter has the lowest recovery error and highest coherence in the experiments for SIR larger than -10 dB.

TABLE I
RECOVERY ERROR, MEAN COHERENCE, AND TIME COST FOR THE EXPERIMENT IN FIG. 12 USING THE HOMOGENEOUS/HETEROGENEOUS SCENES WITH -5-DB SIR

	recovery error	mean coherence	time cost
PCA	-13.81/-1.93 dB	0.974/0.964	0.78 s
RPCA	-15.12/-24.47 dB	0.983/0.988	56.89 s
2-D SPECAN	-23.31/-27.28 dB	0.997/0.996	0.17 s

In addition to recovery accuracy and phase preservation, processing speed is also an important aspect of an RFI filtering algorithm. In our experiments, 2-D SPECAN is the fastest one among the three algorithms. Specifically, the time costs of 2-D SPECAN, PCA, and RPCA for processing the SLC images

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with 2000×2000 pixels are about 0.17, 0.78, and 56.89 s, respectively (using i7-8700 CPU and 32.0-GB DDR4 RAM).

VII. Conclusion

In this article, we investigated the removal of RFI artifacts caused by pulsed LFM signals in SAR SLC images. We started with the signal model of pulsed LFM RFI artifacts in SLC images, and based on the signal model, we pointed out that the RFI artifacts can be focused as sinc-shaped points via matched filtering. Since the original matched filtering is less computationally efficient, we designed a 2-D SPECAN method to replace it, which can efficiently focus RFI artifacts as narrow points in the 2-D spectral domain. For the removal of the focused RFI data points, we designed a 2-D notch filtering method that uses spectrum thresholding and spectral set augmentation to construct notches in the 2-D spectral domain. Since the key point of the proposed method is the 2-D SPECAN operation, we define the proposed RFI filtering kernel as a 2-D SPECAN filter. For algorithm implementation, we provided simplified processing steps and discussed some details, including FM rates' estimation and blockwise processing. Finally, we presented several experiments with real Sentinel-1 SLC images polluted pulsed LFM RFI, which demonstrated the powerful ability of the proposed method.

At the end of this article, we would like to point out the limitations of the proposed method as a parametric approach: it requires the knowledge of azimuth and range FM rates of RFI artifacts. If not known *a priori*, these parameters need to be estimated from data, which may be difficult for weak RFI. Future works will investigate solutions to this problem.

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